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COURSE CODE : DSA0304

COURSE NAME : NATURAL LANGUAGE PROCESSING

Activity 1: Backoff

Using backoff with a penalty of 0.4, find the score for each phrase. Say which level (trigram, bigram or unigram) you used.

a) "the cat sat" (1 mark)

Score: P(sat|the cat) Level: Trigram

b) "dog ate fish" (1 mark)

Score: 0.4 x P(fish|dog ate) if trigram is unseen Level: Bigram backoff

c) "dog sat fish" (1 mark)

Score: Use the highest available n-gram probability, applying a 0.4 penalty for each backoff step. Level: Trigram → Bigram → Unigram, as needed

Activity 2: Deleted interpolation

Part A (2 marks)

Using λ1 = 0.1, λ2 = 0.3, λ3 = 0.6, find P(sat | the cat).

$P(\text{sat}\|\text{the cat}) = 0.1P(\text{sat}) + 0.3P(\text{sat}\|\text{cat}) + 0.6P(\text{sat}\|\text{the cat})$

Part B (4 marks)

Held-out sentence: "the dog ate fish" (trigrams: "the dog ate", "dog ate fish"). Compare Set 1 λ = (0.1, 0.3, 0.6) and Set 2 λ = (0.3, 0.6, 0.1). Find each trigram's interpolated probability and add the natural logs (ln). Which set would deleted interpolation choose, and why?

Set 1: P1=0.1P(ate)+0.3P(ate | dog)+0.6P(ate | the dog) P2=0.1P(fish)+0.3P(fish | ate)+0.6P(fish | dog ate) $L_1 = \ln(P_1) + \ln(P_2)$ Set 2: Q1=0.3P(ate)+0.6P(ate | dog)+0.1P(ate | the dog) Q2=0.3P(fish)+0.6P(fish | ate)+0.1P(fish | dog ate)

L2=ln(Q1)+ln(Q2)

Activity 3: Entropy and perplexity

Part A (2 marks)

Weather: Sunny p = 0.5, Cloudy p = 0.25, Rainy p = 0.25. Find the entropy and the perplexity.

Entropy:

H=-[0.5log2 0.5+0.25log2 0.25+0.25log2 0.25]

Answer: (H=1.5) bits
Perplexity:

PP=2^{1.5}

Answer: (PP approx 2.83)

Part B (4 marks)

List the words that come after "the" and their probabilities. Find the entropy and perplexity of guessing the next word after "the". What is the entropy of the next word after "sat"?

Entropy:
H=-[0.5log2 0.5+0.25log2 0.25+0.25log2 0.25]

Answer: \(H=1.5\) bits

Perplexity:
PP=2^{1.5}

Answer: (PP approx. 2.83)
Entropy after “sat”: H=0 bits, assuming only one possible next word.